

Major Project 21CSA699A

Interim Report

Title: AI-Based Smart Attendance System Using Face Recognition

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Abstract :

Traditional attendance methods take more time and may allow proxy attendance. This project aims to develop a smart attendance system using face recognition technology. The system detects and recognizes a person's face and automatically marks attendance with the current date and time.

Python and OpenCV are used to build the system. Haar Cascade is used for face detection, and the LBPH algorithm is used for face recognition. The attendance details are stored in a CSV file. This system reduces manual work, improves accuracy, and provides a simple and secure method for attendance management in classrooms and educational institutions.

1. Objectives

- ☐ To develop an automatic attendance system using face recognition.
- ☐ To detect and recognize students using a camera.
- ☐ To store attendance details with date and time.
- ☐ To reduce manual work and avoid proxy attendance.
- ☐ To improve the accuracy of attendance management.

2. Scope

This project focuses on creating a smart attendance system for educational institutions. It uses face recognition technology to identify students and mark attendance automatically. The system includes collecting student images, training the model, detecting faces, recognizing students, and storing attendance records in a CSV file. It works best in classrooms with proper lighting and registered students. In the future, the system can be improved by connecting it with databases, mobile applications, or cloud storage. More advanced AI models can also be used to increase accuracy.

3. Introduction

Attendance is an important part of every educational institution. Usually, attendance is taken manually using registers or sheets, which takes time and can lead to mistakes.

Technology has made it possible to automate many daily activities. Face recognition is one such technology that can identify people based on their facial features. It is easy to use and does not require physical contact.

This project uses artificial intelligence and computer vision techniques to create an automatic attendance system. The system captures images, detects faces, recognizes registered students, and marks attendance automatically.

The project helps save time, reduce errors, and prevent proxy attendance. It also provides practical knowledge about image processing, machine learning, and real-time applications.

3.1 Background

Artificial Intelligence (AI) is used in many fields such as healthcare, education, banking, and security. Computer vision is a branch of AI that allows computers to understand images and videos.

Face recognition is a popular application of computer vision. It identifies a person by comparing facial features with stored images. Many organizations use face recognition for security and attendance purposes.

This project uses face recognition to automate the attendance process and make it more reliable and efficient.

3.2 Related Works/Existing System

Traditional attendance systems use paper registers, which are time-consuming and can lead to errors.

Some institutions use RFID cards for attendance. However, students can exchange cards and mark proxy attendance.

Fingerprint systems provide better security, but they require physical contact and special devices.

Face recognition systems are more convenient because they are contactless and automatic. The proposed system uses the LBPH algorithm, which is simple, fast, and suitable for small datasets.

Gap in Existing Systems

Many existing systems need expensive hardware or complex software. This project aims to provide a low-cost and easy-to-use attendance system using basic computer vision techniques.

3.3 Detailed Problem Statement

Manual attendance systems take time and are not always accurate. Students may mark attendance for others, which creates problems in maintaining proper records.

The goal of this project is to develop an automatic attendance system that identifies students through face recognition and records attendance without human intervention.

The system should be simple, accurate, secure, and easy to use in classrooms.

3.4 Methods/Algorithms

Dataset Collection:

Student images are collected and stored in separate folders.

Image Preprocessing:

Images are converted into grayscale format to improve processing speed and accuracy

Face Detection:

The Haar Cascade algorithm is used to detect faces in the images.

Face Recognition:

The LBPH (Local Binary Pattern Histogram) algorithm is used to recognize registered students.

Attendance Recording

When a student is recognized, the system automatically stores the student's name, date, and time in a CSV file.

Tools Used

- Python
- OpenCV
- NumPy
- Pandas
- Google Colab
- Haar Cascade Classifier
- LBPH Face Recognizer

3.5 Expected Results

The system is expected to:

- Detect and recognize students correctly.
- Mark attendance automatically.
- Store attendance records with date and time.
- Reduce manual work for teachers.
- Prevent proxy attendance.
- Save time during class hours.

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